

Use the Django and REST Frameworks to Create a Simple API



This article is a short tutorial on how to create a simple TaskAPI with SQLite, which communicates over JSON and REST. We will use the Django Web and REST frameworks.

Nowadays, we live in a multi-platform world. Everyone has at least one computer, tablet, smartphone or smartwatch. Our little gadgets store and retrieve data. What a great opportunity this presents to touch billions of people with excellent applications, which even those without any computer knowledge can handle.

Whatever front-end you can think of, on whichever platform, you still need a way to store data. The back-end is the heart of every simple and complex application.

In this article, we will discuss how to create a simple TaskAPI with SQLite, which communicates over JSON and REST. We will use the Django Web and REST frameworks.

Virtual environment and project set-up

First of all, it is a good habit to set up a new Python virtual environment to quarantine the requirements from the rest of the system. You can install *virtualenv* and all the other requirements over PyPI.

I assume you have already installed Python 3.X and *virtualenv*.

```
$ mkdir Taskproject
$ cd Taskproject
$ virtualenv env
$ source env/bin/activate
```

On Windows use 'env\Scripts\activate'

Now, we are in our virtual environment and every requirement we want to install will be placed inside the *env* folder. Let's install the necessary requirements in specific versions via Pip.

In this article, we use Python 3.X, Django 1.9 and the Django REST framework 3.0.

```
~# Install Django and Django REST framework into the virtualenv
$ (env) pip install django
$ (env) pip install djangorestframework
# by default PIP install the latest version.
```

Creating a Django project and app

A Django project can manage multiple Django applications. We are going to create a project called *TaskAPI*, and an application called *Task* inside the *TaskAPI* project.

```
$ (env) django-admin.py startproject TaskAPI
$ (env) cd TaskAPI
$ (env) django-admin.py startapp Task
```

Adjusting the project settings

The settings are defined in the file */TaskAPI/settings.py*. First, we have to add the installed apps. For our application, we need to install the Task app along with the mandatory Django application, and the Django REST framework, *rest_framework*.